

Week 12 (CS 418 @ UIC)

Week 12 Slide Deck

Intro to Network Analysis

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Topic Title Placeholder

CS 418 · Week 12 · 🟠 Western 🟠

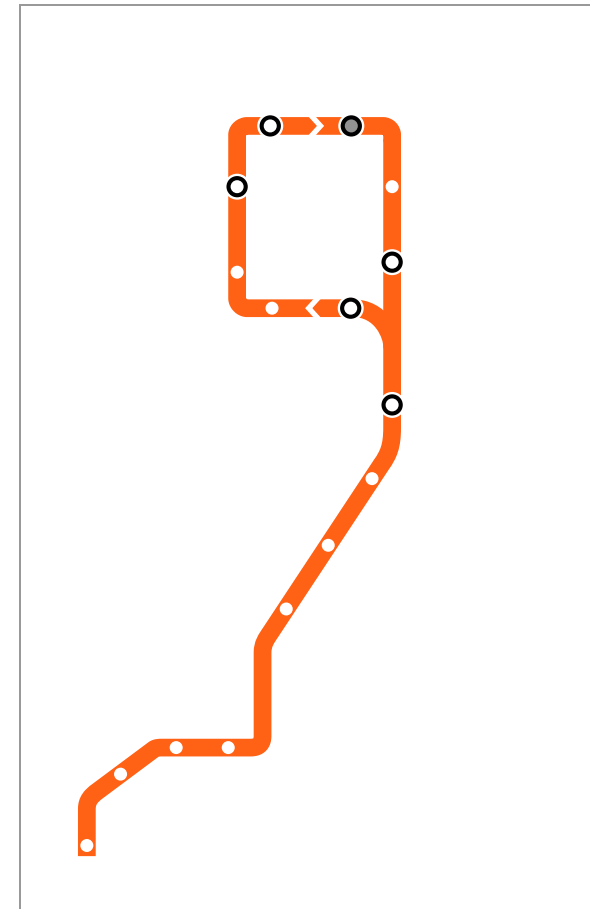




What is a Graph?

A graph represents objects and relationships

- $G = (V, E)$
- V = vertices (objects)
- E = edges (relationships)

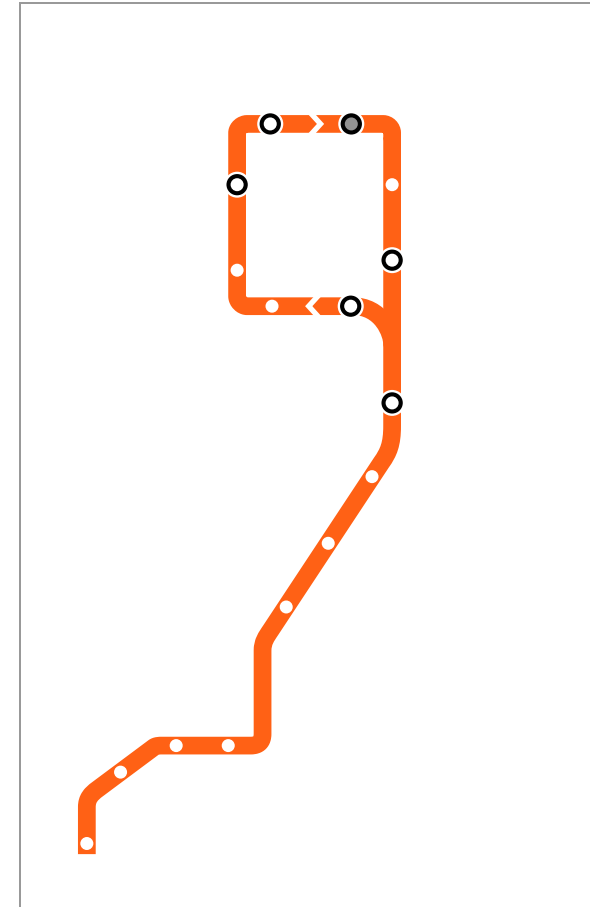


Social network graph (*placeholder*)

Graph Applications

Graphs model many real-world systems:

- World Wide Web
- Social networks
- Transportation networks
- Computer networks

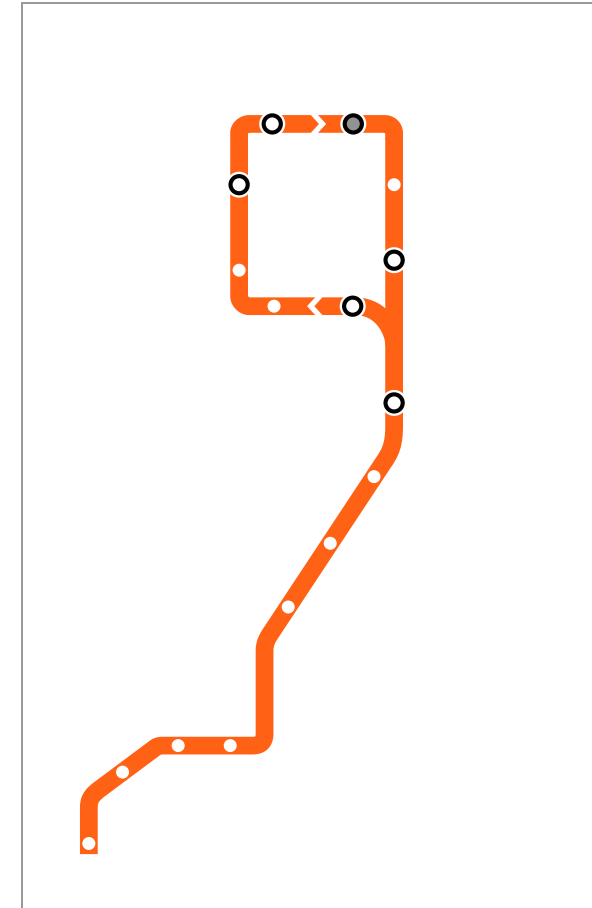


Graph applications (*placeholder*)

Graph Terminology

Key concepts:

- **Endpoints:** Vertices connected by an edge
- **Adjacent:** Vertices connected by an edge
- **Degree:** Number of edges on a vertex
- **Self-loop:** Edge from vertex to itself

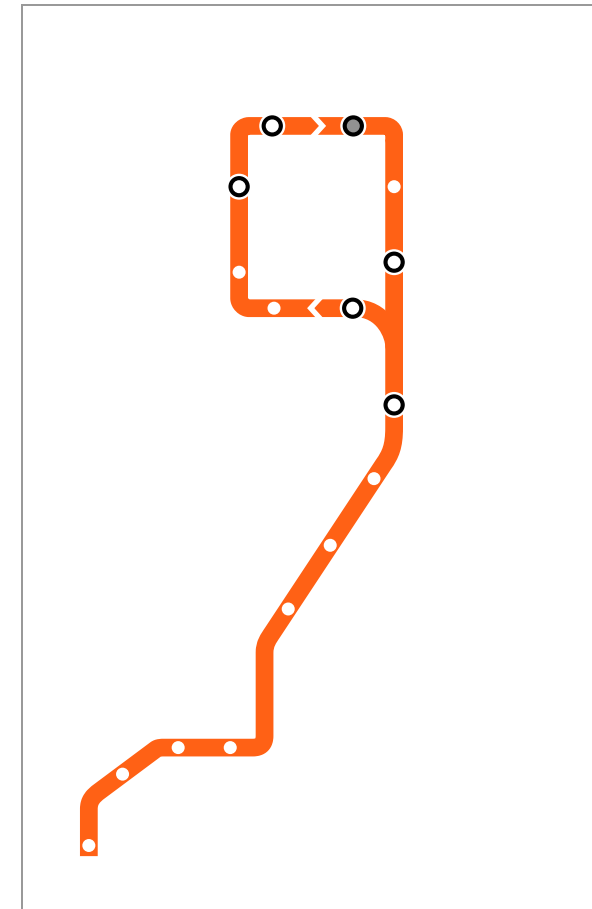


Graph terminology (*placeholder*)

Adjacency Matrix

Matrix representation of graphs

- Efficient for dense graphs
- Easy to check connections

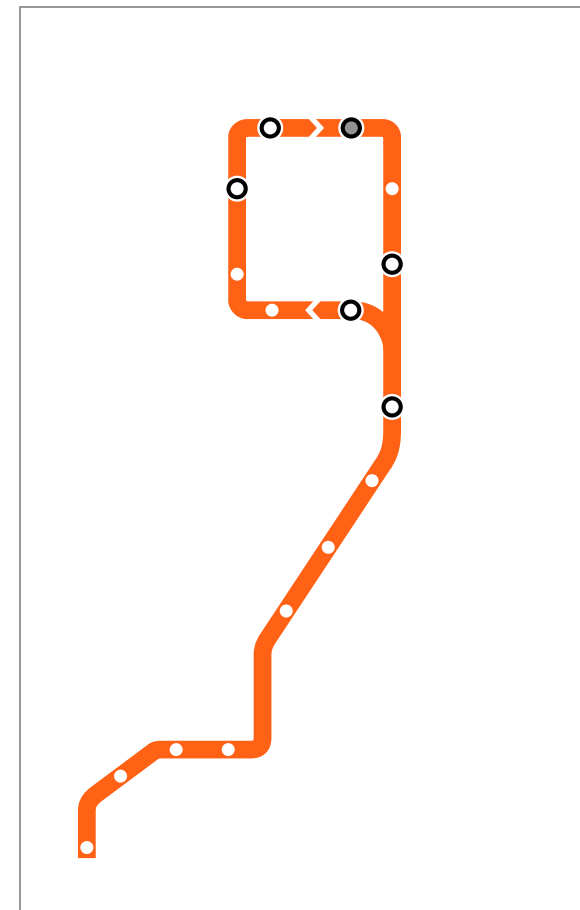


Adjacency matrix example (*placeholder*)

Types of Graphs

Three important types:

- **Directed:** Edges have direction
- **Weighted:** Edges have values
- **Attributed:** Nodes have properties

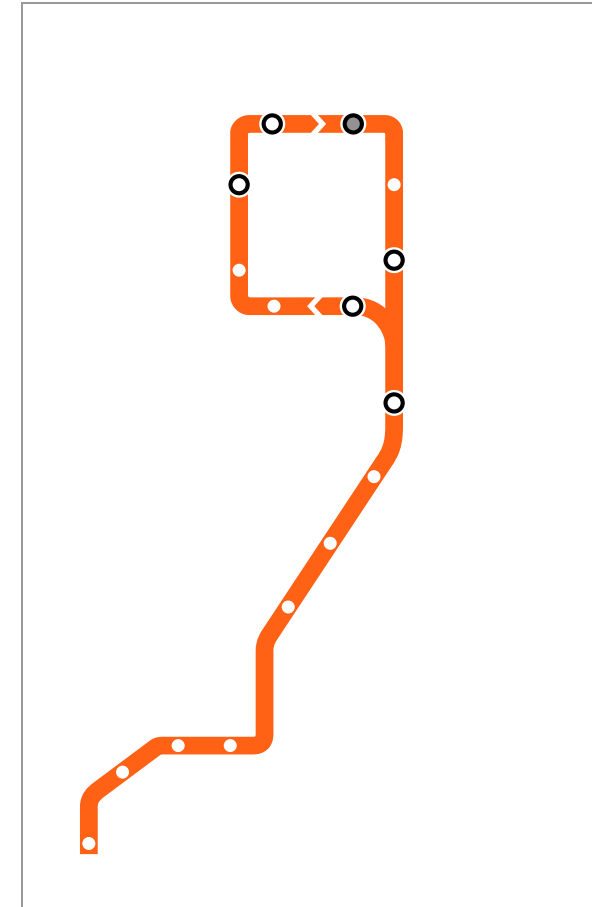


Graph types (*placeholder*)

Community Detection

Finding groups in networks

- Communities = densely connected groups
- Sparse connections between groups
- Social groups in social networks



Community detection (*placeholder*)

Goodness Metrics

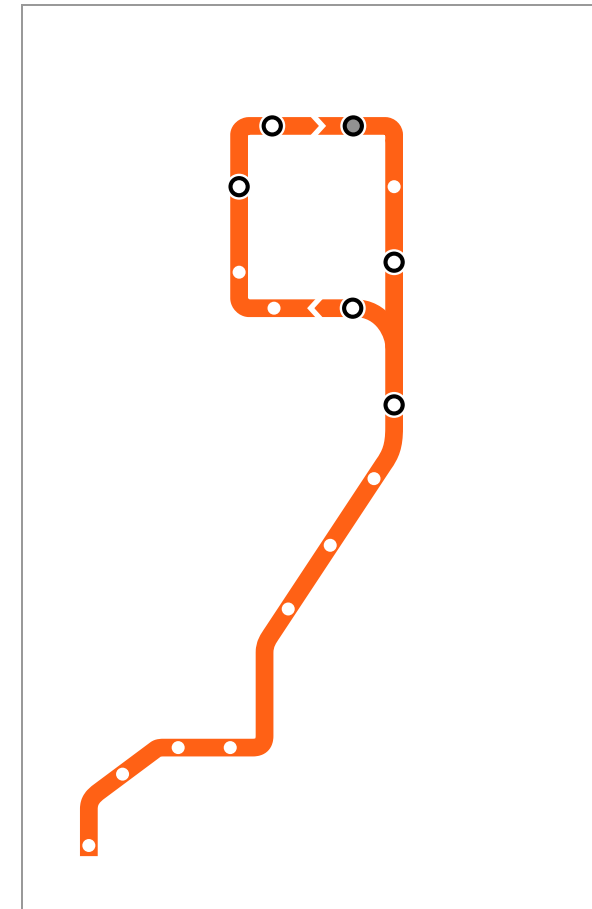
Measuring community quality

Density: Ratio of actual to possible edges

$$\frac{2E_s}{S(S-1)}$$

Conductance: Fraction of edges leaving

$$\frac{O_s}{2E_s + O_s}$$

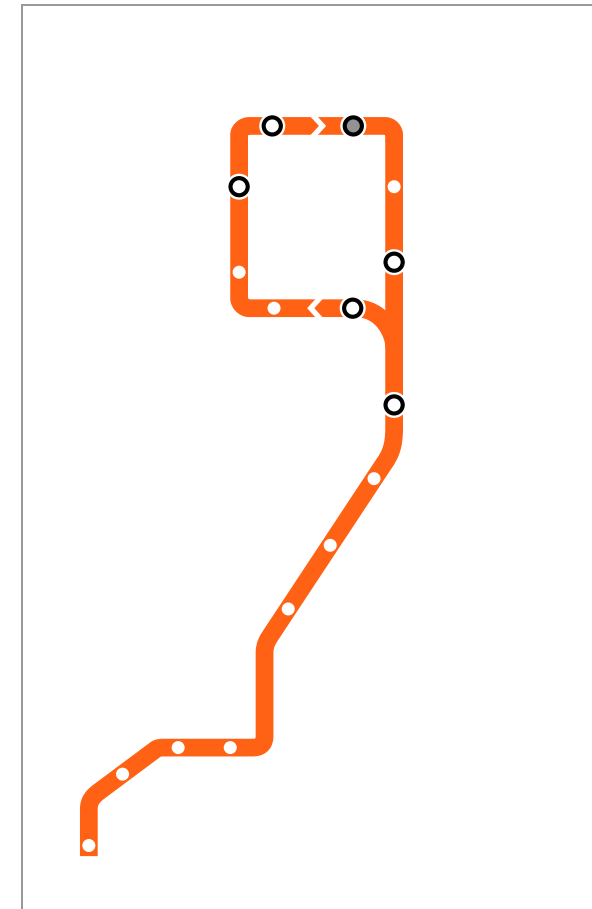


Modularity concept (*placeholder*)

Modularity

Most popular community detection metric

- Actual edges – Expected edges
- Higher Q = better communities
- Maximizing Q is NP-complete

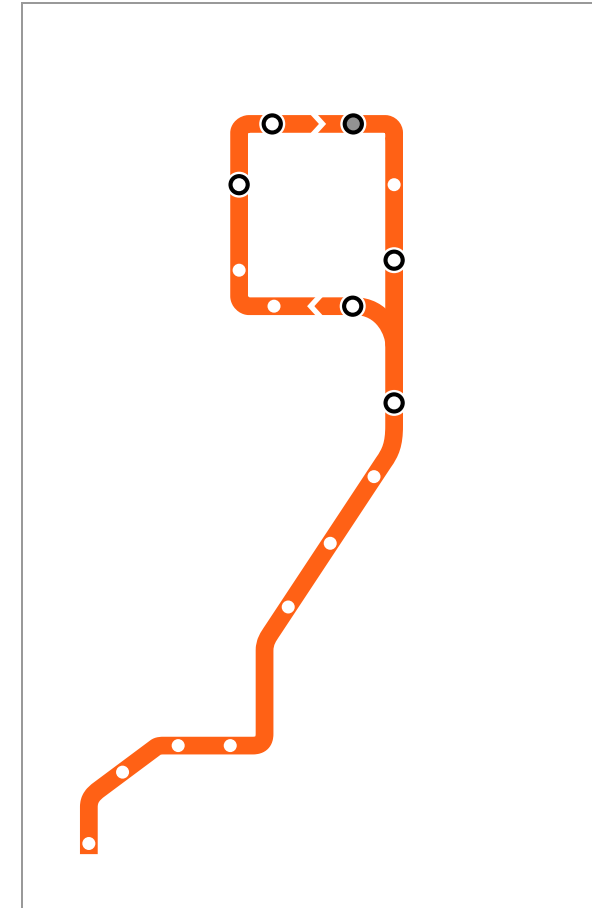


Modularity visualization (*placeholder*)

Louvain Method

Greedy modularity optimization

1. Start with each node as community
2. Move nodes to maximize Q
3. Aggregate communities into supernodes
4. Repeat until convergence



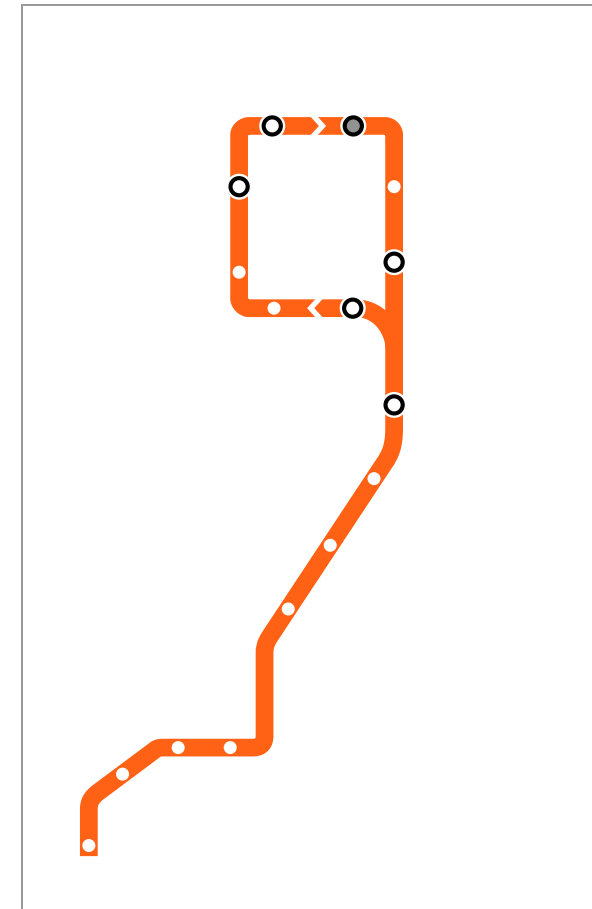
Community detection (*placeholder*)

Network Diffusion

How events spread through networks

Key questions:

- Will it spread?
- Which nodes to target?
- How to stop it?



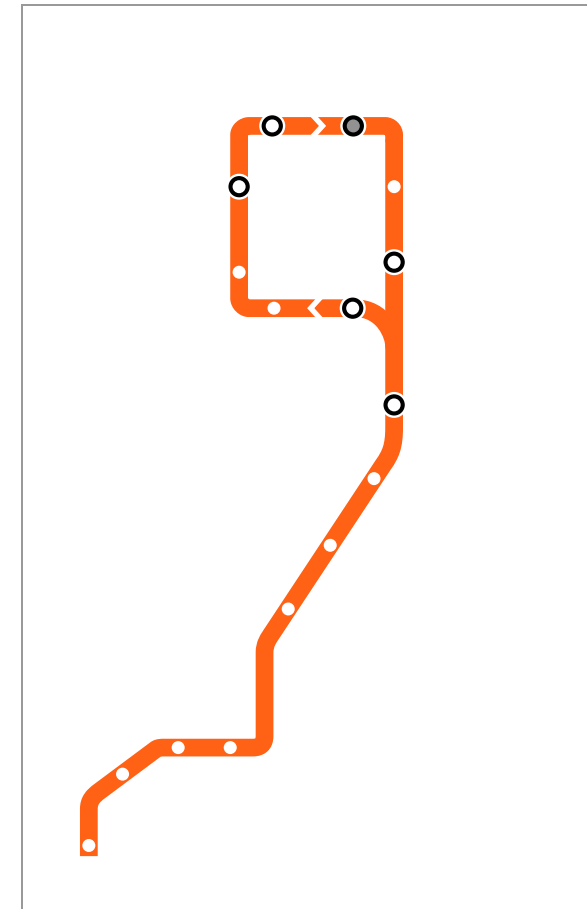
Simple network (*placeholder*)

Broadcast vs. Viral

Two ways content reaches a crowd

(Goel et al. 2016)

- **Broadcast:** one source reaches many directly (e.g. a front page, the Super Bowl)
- **Viral:** person-to-person, over many generations
- Real cascades mix the two



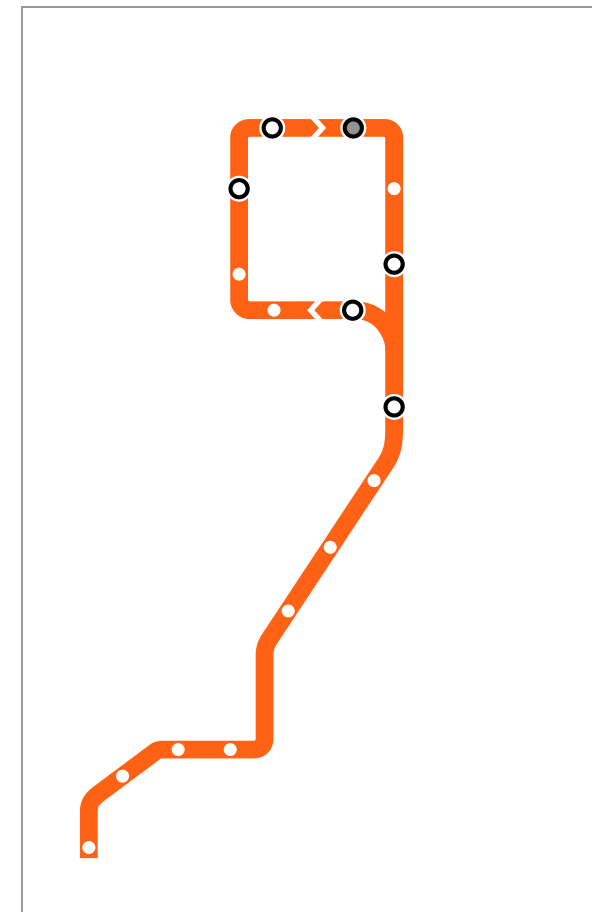
Broadcast (shallow) vs. viral (deep)
diffusion trees (*placeholder*)

Structural Virality

Average distance between all pairs of nodes in a diffusion tree (the *Wiener index*):

$$v(T) = \frac{1}{n(n-1)} \sum_{i=1}^n \sum_{j=1}^n d_{ij}$$

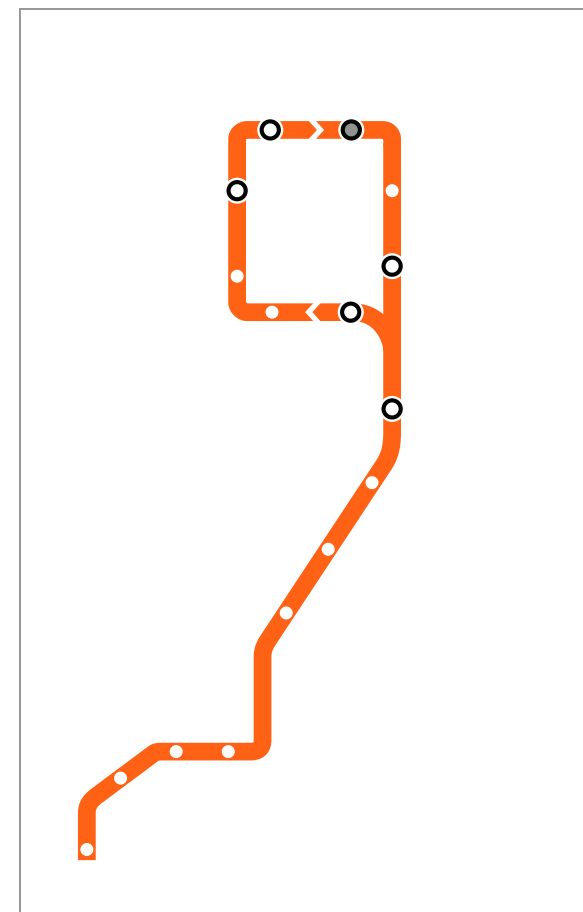
- $\nu(T) \approx 2$: pure broadcast (a star)
- Large $\nu(T)$: deep, multigenerational spread



Diffusion tree with pairwise distances (placeholder)

~1 billion Twitter diffusion events (Goel et al. 2016)

- Over 99% of cascades die in one generation
- Virality varies widely (**no** “tipping point”)
- Popularity is driven mostly by the largest broadcast

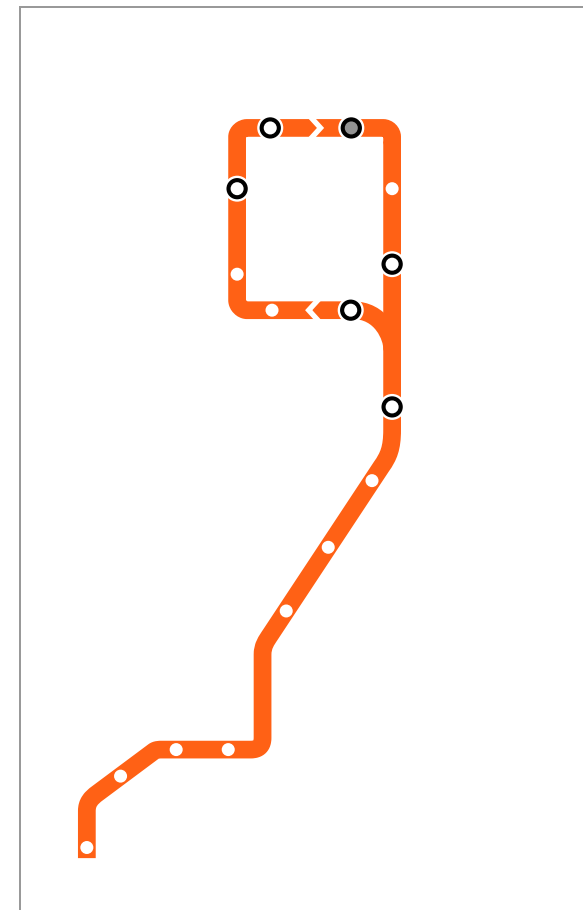


Distribution of cascade sizes and structural virality (*placeholder*)

Virus Propagation Models

VPMs simplify disease spread

- How virulent?
- Recovery rate?
- Immunity?
- Immunity duration?

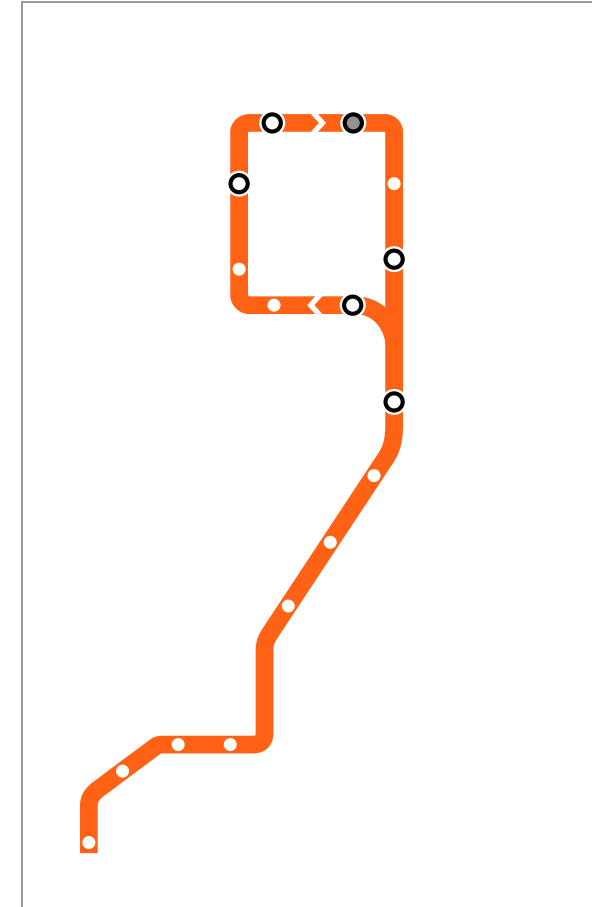


SIS model (*placeholder*)

SIS Model

Susceptible $\rightarrow$ Infected $\rightarrow$ Susceptible

- Example: flu
- β = transmission probability
- δ = healing probability
- No immunity

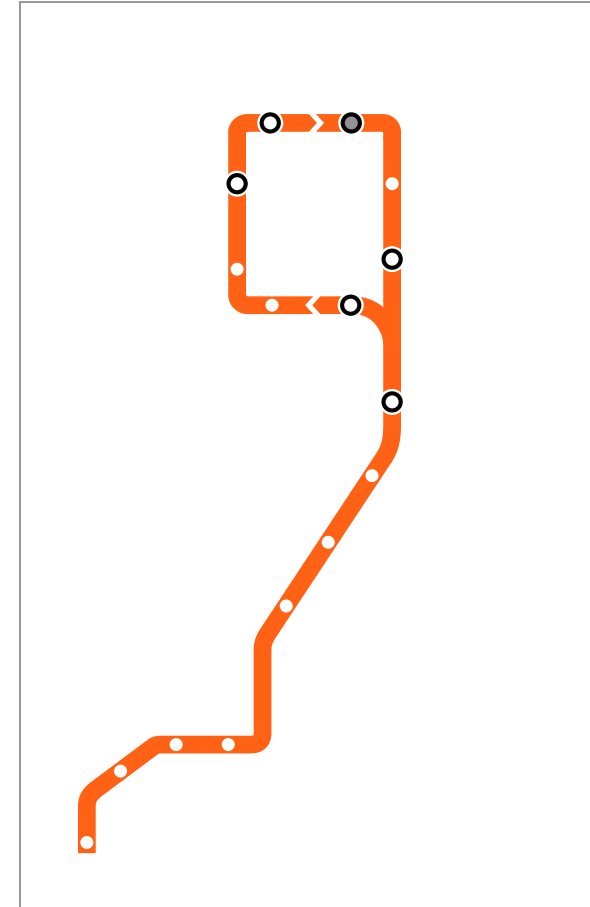


SIS model diagram (*placeholder*)

SIR Model

Susceptible → **Infected** → **Recovered**

- Example: mumps
- β = transmission probability
- δ = healing probability
- Life-time immunity

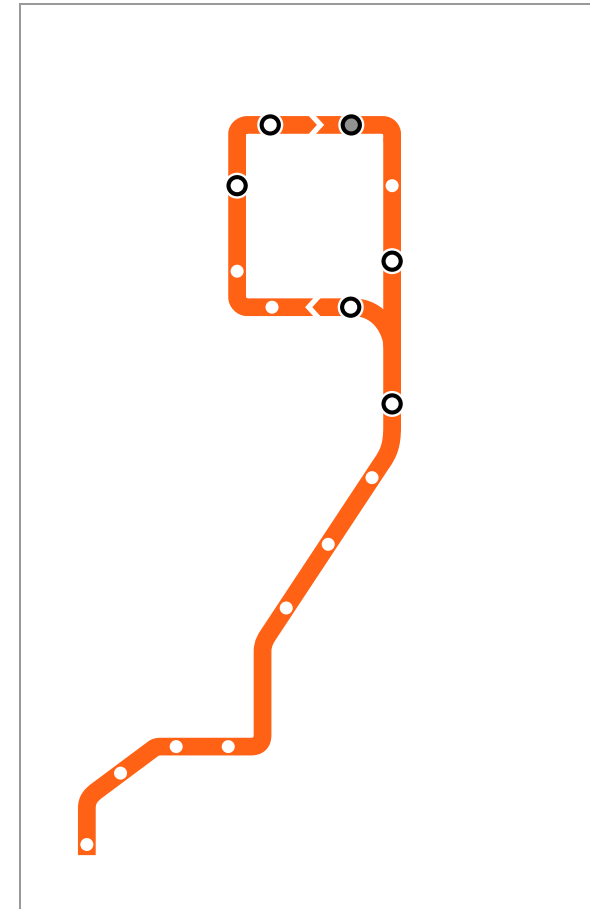


SIR model diagram (*placeholder*)

SIRS Model

Temporary immunity

- Example: whooping cough
- γ = immunity loss probability
- Immunity wears off over time

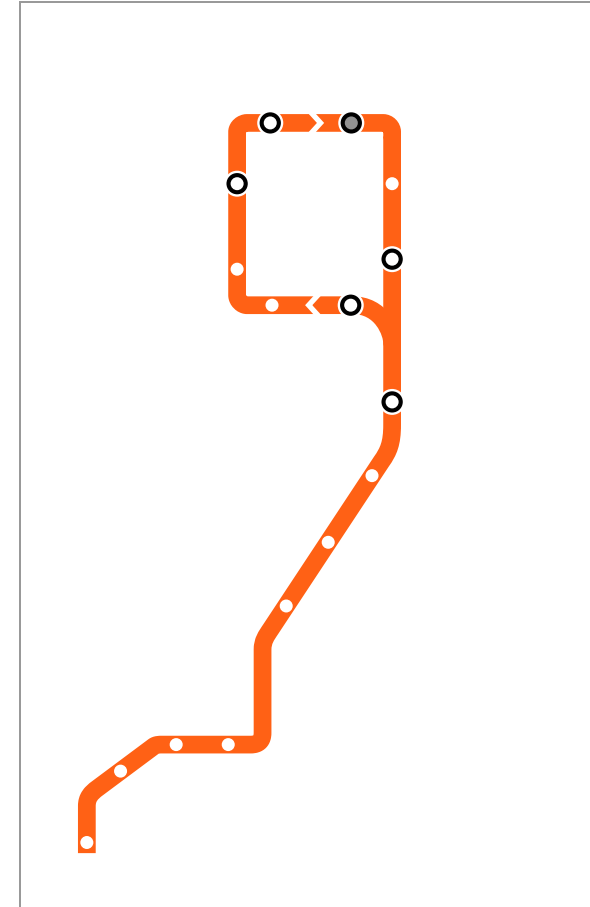


SIRS model diagram (*placeholder*)

SEIV Model

Most complex VPM

- **E** = Exposed (incubation)
- **V** = Vaccinated
- ε = virus maturation
- θ = vaccination rate



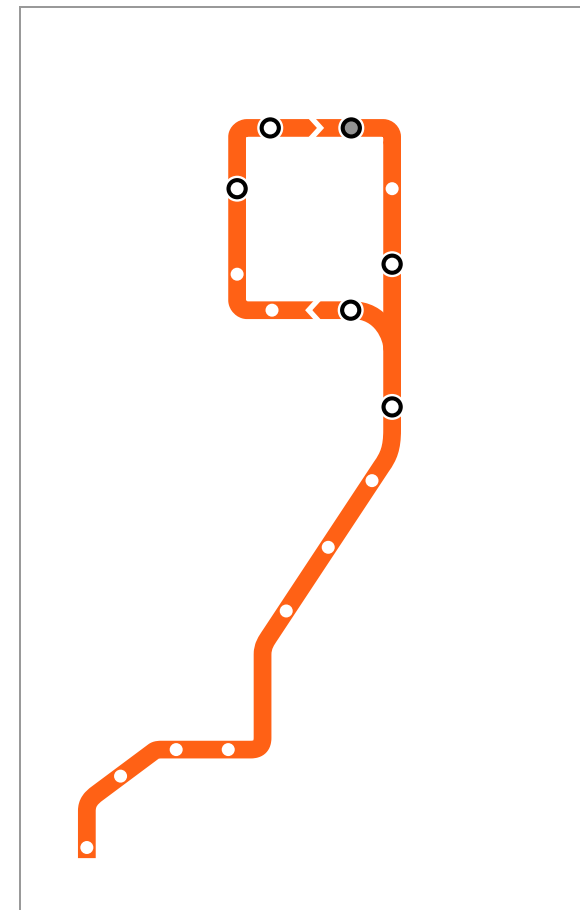
SEIV model diagram (*placeholder*)

Epidemic Threshold

Critical point for virus spread

$$s = \lambda_1 \cdot C_{VPM}$$

- $s < 1$: Virus dies out
- $s > 1$: Epidemic spreads
- λ_1 = network connectivity

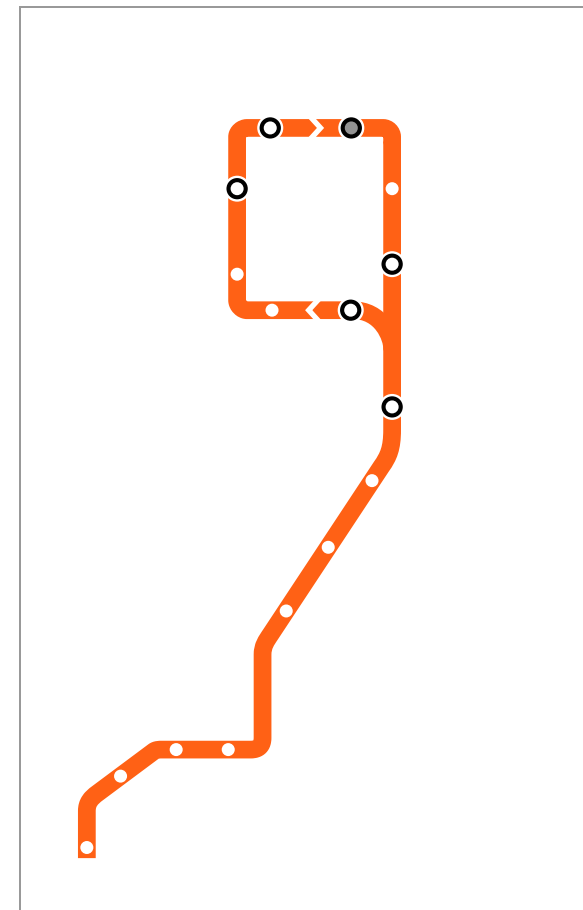


Epidemic threshold (*placeholder*)

Network Connectivity

Why spectral radius matters

- Average degree ignores paths
- λ_1 considers all path lengths
- Better predicts spread



Network connectivity examples
(placeholder)

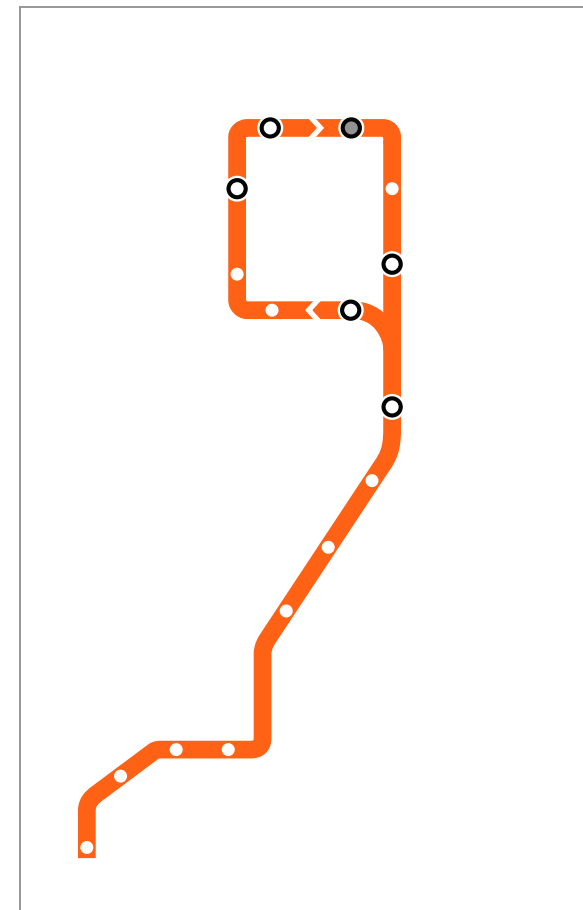
Effective Strength

Predicting epidemic spread

$$s = \lambda_1 \cdot C_{VPM}$$

For SIS: $s = \lambda_1 \cdot \frac{\beta}{\delta}$

- Higher connectivity $\rightarrow$ more spread
- Higher transmission $\rightarrow$ more spread
- Higher healing $\rightarrow$ less spread



Epidemic threshold visualization
(placeholder)

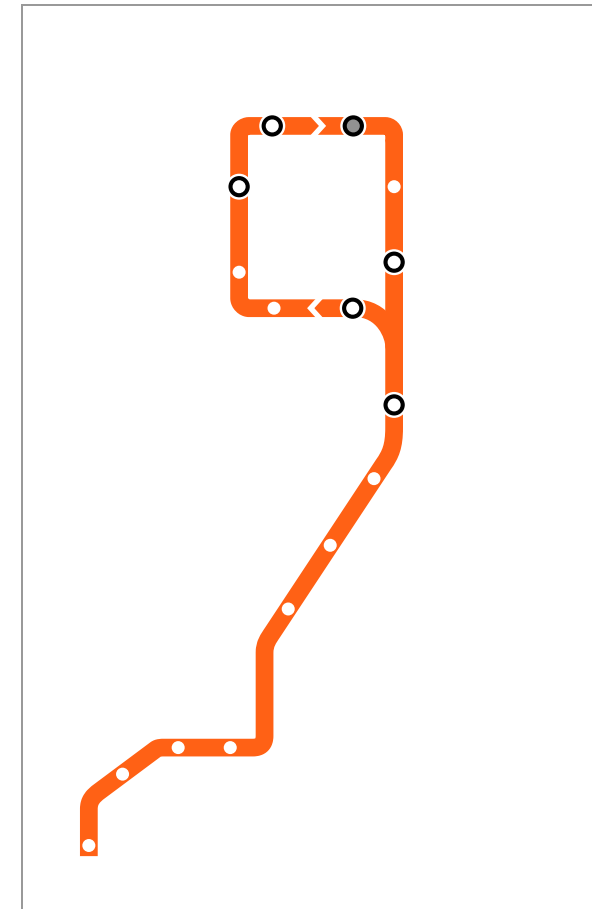
Immunization Problem

Which k nodes to vaccinate?

Goal: Minimize λ_1

Approaches:

- Random (poor)
- High-degree nodes (better)
- NetShield (best)

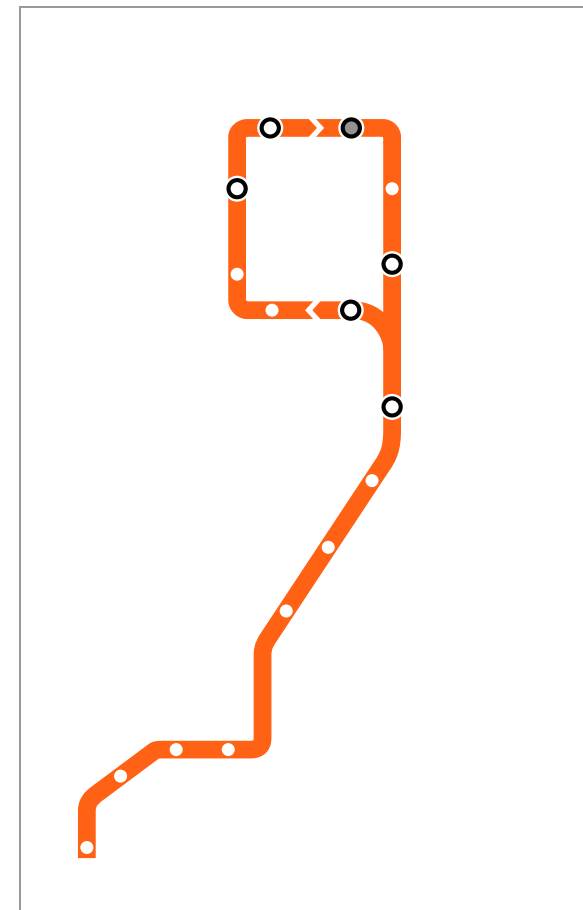


Network connectivity (*placeholder*)

NetShield Algorithm

Efficient immunization

- Approximates eigen-drop
- Greedy selection
- Outperforms heuristics



Community detection showing key nodes (*placeholder*)

References & Credits

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- GitHub source: <https://github.com/jackbandy/data-science-fun/blob/main/docs/slides/week12.md>.
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